

CEO
M. Bilal Javaid

Valid from: 26th January 2026

Valid To: 2nd February 2026

Feasibility Report

LED RGBW Automotive Tail Light



Client: Nicolas

About
M. Bilal Javaid

[Link to Intro](#)

- 8+ years of experience in industrial design and electronics engineering.
- Expertise in PCB design, 3D modeling, and Prototyping.
- Specialized in embedded systems (STM32, ESP32, Raspberry Pi)
- Proficient in CAD, Rendering and DFM
- Builds practical, manufacturable tech solutions by uniting design and engineering.

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Project Scope

This project involves the complete design and development of a **custom automotive-grade RGBW LED tail light control system** integrating PCB hardware, embedded firmware, and Bluetooth-enabled user control. The system will be engineered to support essential vehicle lighting functions including running lights, brake lights, and turn signals using PWM-controlled brightness levels while also enabling RGB lighting effects through a mobile Bluetooth application. The PCB will be designed to withstand automotive electrical conditions such as voltage fluctuations, noise, reverse polarity, and transient surges, ensuring reliable performance in real-world vehicle environments. A priority-based lighting logic system will be implemented where brake signals override turn signals, and turn signals override running lights to ensure compliance with automotive safety standards. The firmware will manage LED driving, Bluetooth communication, lighting modes, and protection mechanisms while maintaining power efficiency. The complete solution will include manufacturable PCB files, embedded software, system documentation, and technical support during prototyping and testing phases, delivering a production-ready LED tail light controller system.

Key Requirements

- Automotive-grade RGBW LED PCB design
- Priority lighting logic (Brake > Turn > Running)
- PWM-based brightness control
- Bluetooth RGB control via free mobile app
- Integrated MCU with BLE capability
- Reverse polarity, surge, and noise protection
- Multi-input lighting signal support
- Firmware-controlled lighting modes
- Automotive voltage compatibility (9–16V typical)
- Complete manufacturing documentation

We will deliver a fully functional PCB design, firmware, and documentation package, along with testing support and revision assistance to ensure successful deployment.



Chapter : 01

Research and Development

1.1 Approach

The research and development approach for the RGBW Automotive Tail Light Control System focuses on delivering a high-reliability electronic and embedded software solution, integrating automotive-grade PCB design, real-time firmware development, Bluetooth communication, and system-level validation. The process begins with translating client functional requirements into engineering specifications, ensuring accurate PWM-based brightness control, priority-based lighting logic (Brake, Turn, Running), and robust Bluetooth-enabled RGB control. Automotive electrical constraints such as voltage transients, EMI, reverse polarity, and temperature variations are addressed through protection circuit design, component selection, and layout strategies to ensure long-term operational stability.

The R&D methodology follows a structured design–verify–optimize workflow, where schematic design, PCB layout development, firmware architecture, and system integration are executed in parallel to improve iteration efficiency and reduce technical risk. Firmware development focuses on stable lighting behavior, safety override mechanisms, BLE communication reliability, and fault handling. The PCB design emphasizes signal integrity, thermal management, and manufacturability without including fabrication activities. Extensive simulation, design reviews, bench validation, and functional testing ensure the final design is robust, scalable, and ready for manufacturing deployment.

Phase 1: Requirement Analysis and System Architecture Definition

This phase establishes the technical foundation by translating client requirements into detailed system specifications and functional architecture. Lighting behaviors, brightness levels, priority logic, Bluetooth control modes, and safety constraints are analyzed to define system workflows, power architecture, and communication interfaces. Automotive electrical conditions are evaluated to define protection and reliability requirements.

- Analyze running, brake, turn, and RGB enable input signals
- Define lighting priority logic (Brake > Turn > Running)
- Establish PWM resolution and brightness accuracy requirements
- Define Bluetooth communication behavior and mobile app control
- Develop system block diagram and control architecture

Phase 2: Component Selection and Electrical Design Planning

This phase focuses on selecting suitable automotive-grade components for MCU, LED drivers, power management, and protection circuitry. Multiple alternatives are shortlisted to ensure availability and supply-chain resilience. Power budgeting, thermal feasibility, and current capacity analysis are performed to ensure stable operation under sustained lighting loads.

- Select MCU with BLE (ESP32 / nRF52 families)
- Choose RGBW LEDs and LED driver topology
- Select voltage regulators and protection components
- Perform power consumption and thermal analysis
- Finalize preliminary BOM and component strategy

Phase 3: Schematic Design and Electrical Validation

This phase involves developing the complete electrical schematic including power regulation, MCU core circuits, LED drivers, signal conditioning, Bluetooth RF interfaces, and protection networks. Electrical rule checks and design reviews are conducted to validate voltage margins, current handling, and system stability.

- Design automotive power protection and regulation circuits
- Implement MCU core, RF matching, and programming interfaces
- Design PWM LED driver and current regulation circuits
- Add input signal conditioning for brake, turn, and running inputs
- Perform schematic verification and validation

Phase 4: PCB Layout Design and Design-for-Manufacturing (DFM)

This phase converts the validated schematic into a high-reliability PCB layout optimized for signal integrity, EMI suppression, thermal performance, and manufacturability. Automotive layout practices are applied including optimized grounding strategies, RF layout constraints, and high-current routing techniques.

- Define PCB stack-up and layer configuration
- Optimize component placement for EMI and thermal performance
- Route high-current LED paths and sensitive signal traces
- Apply RF layout constraints for Bluetooth antenna performance
- Perform DRC and DFM validation checks

Phase 5: Firmware Architecture Design and Embedded Software Development

This phase focuses on developing embedded firmware to manage lighting logic, PWM brightness control, Bluetooth communication, and safety fallback behavior. Modular firmware architecture ensures maintainability, scalability, and reliable real-time operation.

- Develop firmware architecture and task scheduling
- Implement PWM-based brightness control algorithms
- Code lighting priority logic (Brake > Turn > Running)
- Implement BLE services and command handling
- Add fault detection and safe fallback operating modes

Phase 6: System Integration and Functional Testing

This phase validates the integrated hardware and firmware design through bench-level testing and simulated automotive operating conditions. Lighting transitions, brightness accuracy, Bluetooth reliability, protection circuit behavior, and fail-safe operation are verified against requirements.

- Validate lighting mode transitions and priority logic
- Test PWM brightness accuracy under voltage variation
- Verify BLE connectivity stability and responsiveness
- Validate protection circuit behavior and fault handling
- Conduct system reliability and stress testing

Phase 7: Design Optimization, Documentation, and Final Validation

The final phase incorporates feedback from system testing into optimized schematic, layout, and firmware revisions. Manufacturing-ready documentation is prepared including design files, firmware binaries, wiring diagrams, and operation manuals. Final system validation confirms readiness for deployment.

- Implement hardware and firmware refinements
- Generate complete design documentation packages
- Prepare firmware release and programming procedures
- Produce wiring and system operation manuals
- Perform final system acceptance validation

Component In-House Inventory

Sr. No.	Component Name	Preferred / Alternative Components	Availability	Purpose
1	MCU + Bluetooth Module	ESP32-WROOM-32 / nRF52840 / ESP32-S3	Available	Main controller and Bluetooth communication
2	RGBW LEDs	SK6812 RGBW / APA102 RGBW / WS2815 RGBW	Available	Tail light illumination and color control
3	LED Driver IC	TLC5947 / CAT4101 / MAX7219	Available	Constant current LED brightness control
4	Power MOSFET	AO3400A / IRLML6344 / Si2302	Available	High-current LED switching
5	Voltage Regulator (Buck Converter)	LM2596 / TPS54202 / MP1584	Available	Step-down voltage regulation (12V → 5V/3.3V)
6	Reverse Polarity Protection	SS34 Schottky Diode / PMOS Reverse Protection / SMBJ Diode	Available	Protects against incorrect power polarity
7	Transient Voltage Suppressor (TVS)	SMBJ36A / SM8S36A / SMAJ36A	Available	Protects against voltage spikes and load dump
8	Input Signal Conditioning Op-Amp	LMV358 / MCP6002 / TLV2372	Available	Buffers brake, turn, and running signals
9	Current Sense Resistor	0.1Ω 1% / 0.05Ω 1% / 0.2Ω 1%	Available	LED current sensing and protection
10	Bluetooth Antenna	PCB Trace Antenna / IPEX Antenna / Chip Antenna	Available	Wireless communication
11	Crystal Oscillator	40 MHz Crystal / 26 MHz Crystal / Internal RC Oscillator	Available	MCU clock stability
12	Power Connector	JST-VH Connector / Molex Micro-Fit / Screw Terminal	Available	Power and signal interfacing
13	Programming Interface	CP2102 / CH340C / FT232RL	Available	Firmware flashing and debugging
14	PCB Substrate	FR4 TG150 / FR4 TG170 / Aluminum Core PCB	Available	Mechanical and thermal reliability

Milestones and Deliverables

Milestones

- System requirements and architecture finalized
- Component selection and BOM approval completed
- Electrical schematic design approved
- PCB layout finalized and DFM checks passed
- Firmware core functionality completed
- Bluetooth communication successfully integrated
- System-level functional testing passed
- Final design documentation approved

Deliverables

- Electrical schematic design files
- PCB layout design files (Gerber, BOM, Pick & Place)
- Embedded firmware source code and binaries
- Bluetooth communication interface implementation
- Wiring diagrams and installation documentation
- System operation and maintenance manual
- Test reports and validation documentation

Timeline

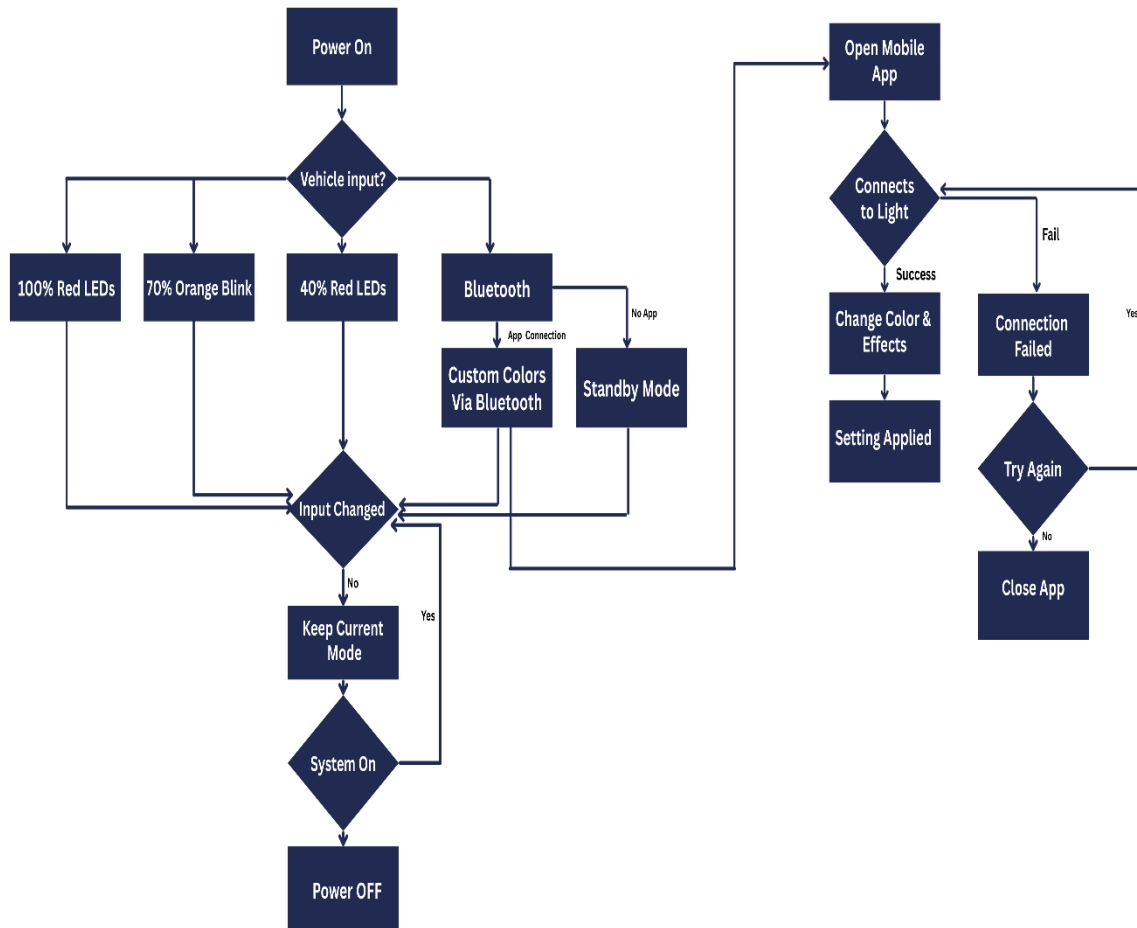
- Requirement Analysis & System Architecture:12 hours
- Component Selection & Electrical Design Planning:14 hours
- Schematic Design & Electrical Validation:24 hours
- PCB Layout Design & DFM Validation:26 hours
- Firmware Architecture & Embedded Software Development:28 hours
- System Integration & Functional Testing:10 hours
- Design Optimization & Documentation:6 hours

Note

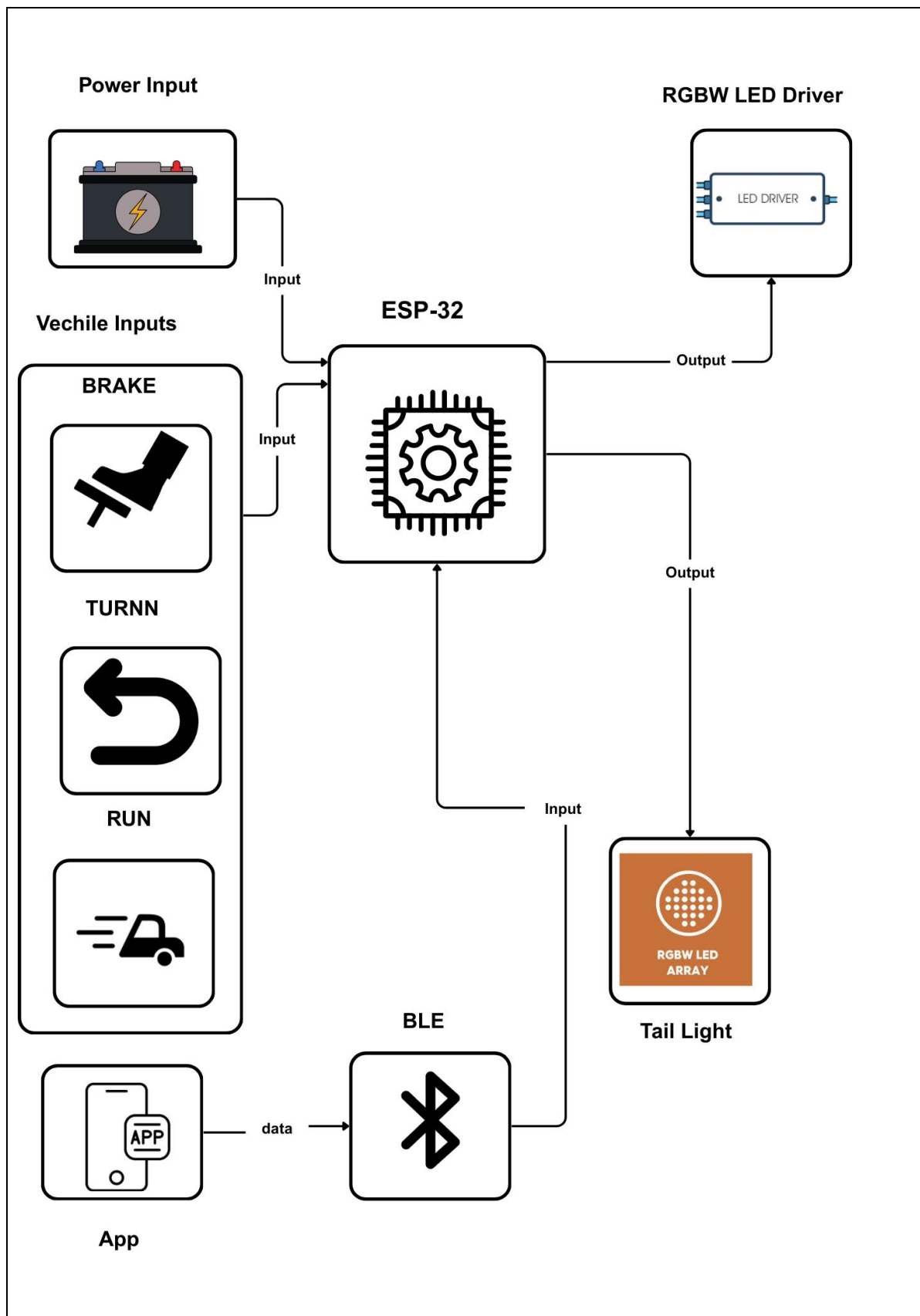
- Total R&D Time: 120 Hours

Total timeline for R&D: 120 Hours

Flow Chart



Block Diagram



Outcomes

- Fully functional automotive-grade RGBW tail light controller
- Stable priority-based lighting control system
- Bluetooth-enabled RGB lighting interface
- Manufacturable PCB design ready for production
- Tested and validated embedded firmware solution
- Reliable integration-ready automotive lighting module



Chapter : 02

PCB Design



2.1 Approach

The PCB design approach for the RGBW Automotive Tail Light Control System focuses exclusively on delivering a robust, automotive-grade electronic design that is electrically reliable, thermally efficient, and manufacturing-ready, without including fabrication activities within the R&D scope. The process begins with transforming system-level functional requirements into detailed schematics, followed by high-quality PCB layout development that ensures safe power handling, stable LED current delivery, accurate PWM control, and reliable Bluetooth communication. Automotive design constraints such as voltage transients, EMI exposure, vibration tolerance, and temperature variation are addressed through appropriate circuit protection, grounding strategies, and layout optimization techniques.

The PCB design workflow emphasizes Design for Manufacturing (DFM) and Design for Assembly (DFA) best practices to ensure the output design files are fabrication-ready while remaining cost-efficient and scalable for production. Special focus is placed on separating high-current LED power paths from low-voltage digital circuitry, optimizing RF layout for Bluetooth performance, and managing thermal dissipation from LED drivers and regulators. Multiple design reviews, simulations, and verification checks are conducted to ensure the PCB meets functional, electrical, and automotive reliability requirements prior to release.

Schematics

Power Input and Protection Circuitry

This sub-section defines the electrical front-end responsible for safely interfacing with the vehicle power system. The design protects downstream electronics from reverse polarity, voltage spikes, and electromagnetic interference while ensuring stable regulated outputs for logic and LED domains.

- Reverse polarity protection using diode or PMOS topology
- Transient voltage suppression for load dump and surge events
- EMI filtering using LC networks and bulk decoupling
- Buck regulation for stepping down automotive voltage to logic rails

MCU and Bluetooth Core Circuitry

This sub-section covers the design of the main control unit using an MCU with integrated Bluetooth functionality. The circuitry ensures stable operation, reliable RF performance, and easy programming and debugging access.

- ESP32 / nRF52 core circuitry and clocking
- Reset, boot mode, and programming interface design
- RF antenna matching network and layout constraints
- Decoupling and grounding for noise immunity

LED Driver and PWM Control Circuits

This sub-section defines the LED driving architecture required to achieve accurate brightness control for RGBW LEDs while maintaining thermal safety and electrical efficiency. PWM signals from the MCU regulate brightness levels and lighting transitions.

- Constant-current LED driver ICs or MOSFET-based drivers
- PWM signal routing and brightness control resolution
- Current sensing and thermal protection mechanisms
- Load balancing for uniform LED illumination

Input Signal Conditioning Circuits

This sub-section addresses conditioning of automotive input signals such as brake, turn, running, and RGB enable inputs. These signals are filtered and level-shifted to ensure compatibility with MCU logic levels and noise immunity.

- Voltage dividers and transient protection
- Op-amp or comparator buffering
- Noise filtering and debounce logic
- ESD protection and fault isolation

Communication and Debug Interfaces

This sub-section includes interfaces for firmware flashing, debugging, and validation, ensuring safe and convenient access during development and testing.

- USB-to-UART or SWD/JTAG programming interface
- Test points for critical signals and power rails
- Safe isolation between external and internal signals

PCB Design

PCB Stack-Up and Layer Planning

This sub-section defines the PCB structure to ensure proper power distribution, EMI suppression, and thermal management. The stack-up is optimized for current handling and signal integrity.

- Multi-layer FR4 PCB selection
- Dedicated power and ground planes
- Copper thickness optimization for LED current paths
- Thermal spreading through copper pours and vias

Component Placement Strategy

Component placement is optimized to minimize noise coupling, improve thermal dissipation, and ensure manufacturability. Power components, RF sections, and digital control circuits are strategically separated.

- Isolation of high-power and low-noise sections
- Optimal antenna placement with RF keep-out zones
- Connector alignment for mechanical integration
- Thermal-aware positioning of regulators and drivers

High-Current and Power Routing

This sub-section focuses on routing techniques for LED power and supply lines to minimize voltage drop, heating, and EMI generation.

- Wide traces and copper pours for LED current paths
- Multiple vias for current sharing and thermal relief
- Minimized loop areas to reduce noise coupling
- Proper return path continuity

Signal Integrity and EMI Control

Sensitive signals and RF traces are routed to preserve signal quality and suppress electromagnetic interference, ensuring reliable operation in automotive environments.

- Controlled impedance routing for RF signals
- Solid reference ground planes
- Stitching vias and shielding structures
- Separation of noisy and sensitive nets

Design for Manufacturing and Assembly (DFM/DFA)

This sub-section ensures the PCB layout complies with fabrication and assembly constraints while remaining scalable for mass production.

- Standard footprints and pad geometries
- Adequate solder mask openings and silkscreen clarity
- Panelization and fiducial placement considerations
- Automated assembly compatibility

Milestones and Deliverables

Milestones

- PCB design requirements finalized
- Electrical schematic completed and reviewed
- Schematic validation and approval
- PCB layout completed
- Design rule and DFM checks passed
- Final design files released for manufacturing

Deliverables

- Complete electrical schematic files
- PCB layout design files
- Gerber fabrication files
- Bill of Materials (BOM)
- Pick-and-place files
- Design verification documentation

Timeline

- Requirement Review & Design Planning – 6 hours
- Component Selection & Library Preparation – 8 hours
- Schematic Design & Electrical Validation – 18 hours
- PCB Layout Design & DFM Checks – 20 hours
- Design Review, Optimization & File Finalization – 8 hours

Note

- Total PCB Design Time: 60 Hours

Total timeline for PCB Team: 60 Hours

Note:

PCB fabrication is not included in the project scope; however, for budgeting purposes, the estimated fabrication cost is **\$100–\$150 per PCB**. For an initial batch of **5 PCBs**, the estimated total fabrication cost ranges between **\$500 and \$750**, excluding assembly and component costs.

PCB fabrication will be carried out through JLCPCB, ensuring high-quality manufacturing and adherence to design specifications. The estimated lead time for fabrication and delivery is approximately 2-3 weeks.

Outcomes

- Automotive-grade, fabrication-ready PCB design
- Stable power distribution and LED driving performance
- Reliable Bluetooth RF layout and communication stability
- Accurate PWM-based brightness control implementation
- EMI-optimized and thermally efficient board layout
- Production-scalable electronic design solution



Chapter : 03

Mechanical Design

3.1 Approach

The mechanical work will begin by understanding the tail light housing constraints and defining a mounting strategy for the PCB and LED array inside the existing lens/body. The focus will be on uniform light diffusion, thermal reliability, and automotive vibration resistance. Once the concept is finalized, a detailed 3D CAD model will be developed for PCB mounting features, LED alignment, diffuser positioning, sealing strategy, and connectors routing. Rapid prototypes, 3D prints and laser-cut diffuser samples will be used to validate fit, optics, and assembly before finalizing production-ready mechanical files.

Important Design Aspects

The design will ensure uniform light output using correct LED spacing, controlled LED-to-lens distance, and diffuser, with separated zones to avoid color bleed. Heat will be managed using MCPCB or thermal coupling to a metal frame to protect the lens and improve reliability.

Components Layout

- LED PCB positioning relative to lens to meet photometric spread
- Dedicated light chambers / separators for:
 - Running (red 40%)
 - Brake (red 100%)
 - Turn (orange 70%)(Right & Left side Led respectively)
- Mount points for:
 - PCB standoffs and fasteners
 - Diffuser or light guide plate
 - Connector bracket and wire routing channel

Material Selection

Prototype:

- PLA / PETG for fit verification and mounting brackets
- Diffuser trials using acrylic/frosted polycarbonate sheets

Production (Typical Automotive):

- PC/ABS blend for internal brackets and mounts
- Polycarbonate lens-compatible diffuser

Mechanical & Component Mounts

- Integrated **standoff bosses** (heat-resistant spacing) for PCB
- Option for **captive screws** or clip-lock retention depending on housing space
- Strain relief clamp for harness exit
- Diffuser/light guide retention using ribs or snap features

Positioning & Integration

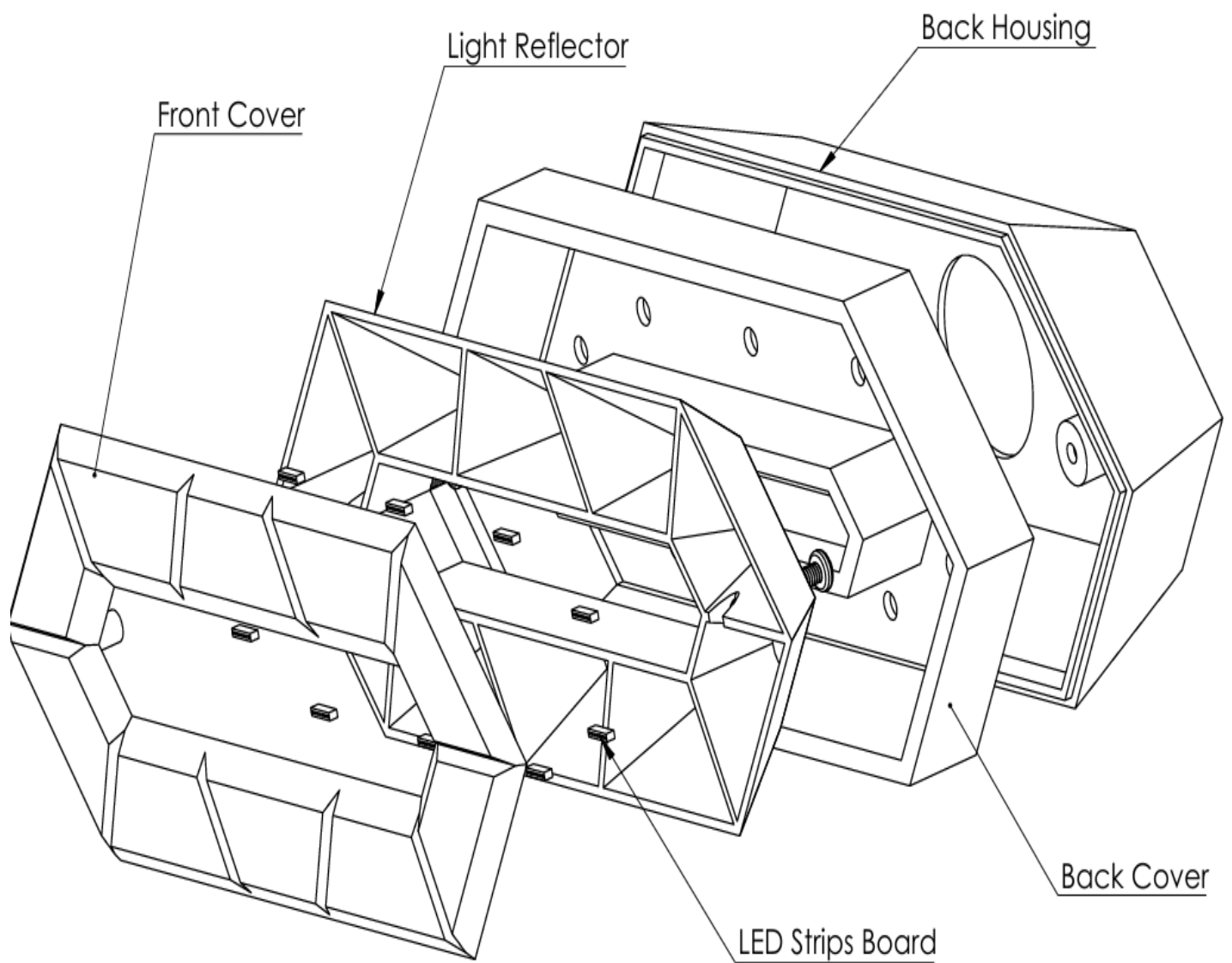
The PCB and diffuser will be positioned to match the **existing reflector/lens geometry** and ensure proper light projection angles. The design will prioritize:

Prototyping & Testing

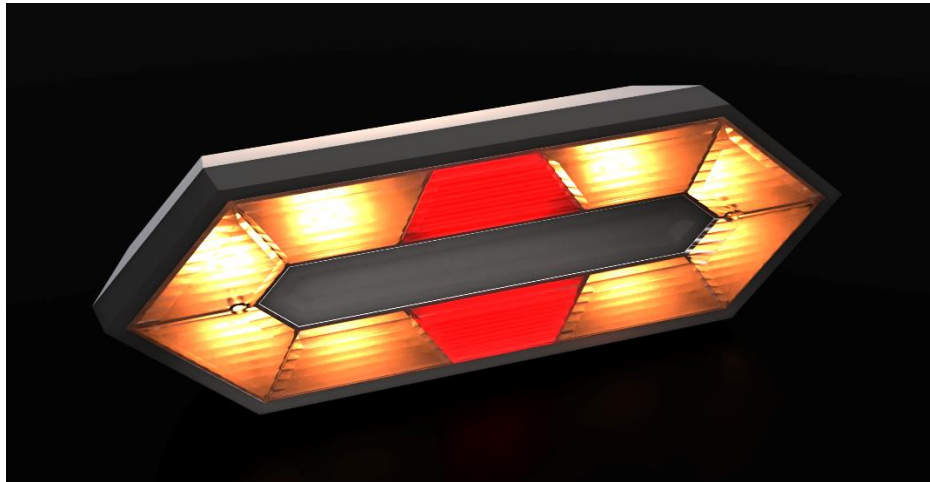
Prototyping:

- 3D printed internal brackets + PCB mock placement
- Diffuser/light guide testing samples
- Fitment tests inside housing

Conceptual Design



Realistic Renders



Milestones and Deliverables

Milestones

- Housing study + mounting concept
- CAD design of internal mounting + diffuser strategy
- Prototype brackets + fit verification
- Optics refinement and sealing refinement
- Final production-ready mechanical package

Deliverables

- 3D CAD files (STEP) for mounts/brackets/diffuser concepts
- STL files for prototyping
- Assembly drawings + exploded view
- Mechanical BOM (fasteners, gaskets, brackets, diffuser materials)

Timeline

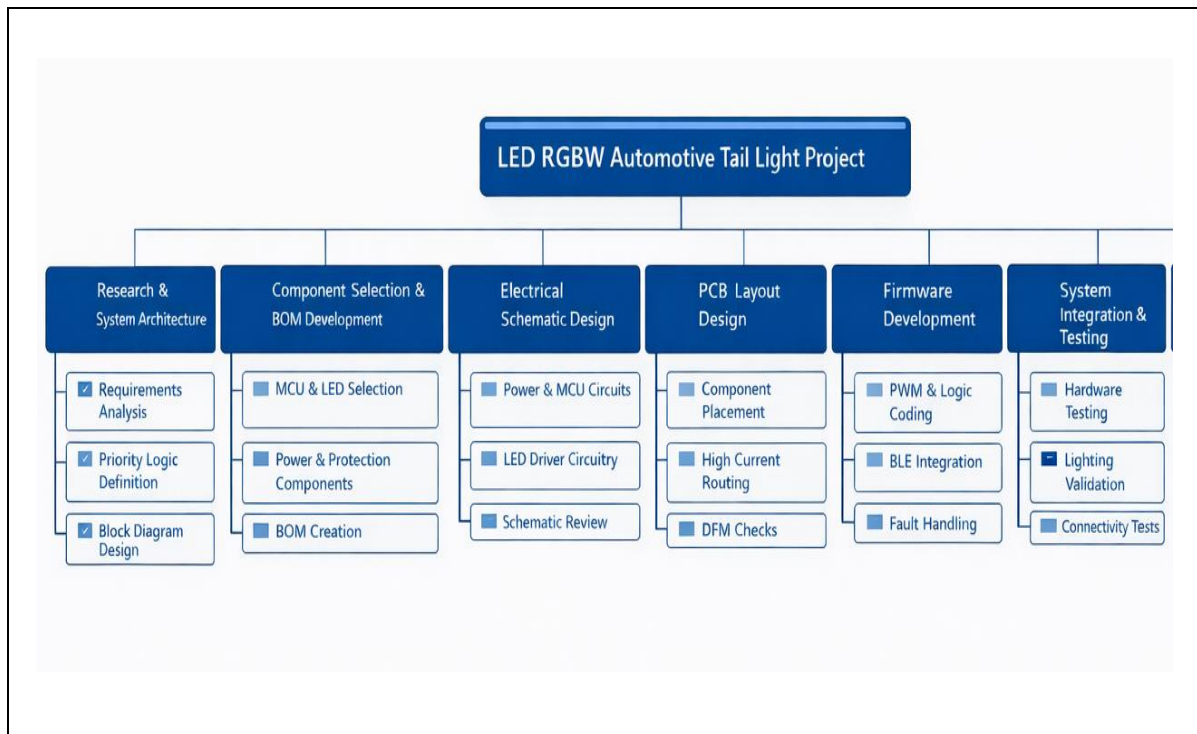
- **Concept Design (Mechanical):** 15 hours
- **Final CAD Design (Mechanical):** 24 hours
- **Prototyping & Testing (Mechanical):** 30 hours
- **Documentation :** 6 hours

Note

- Total Mechanical Design Time: 75 Hours

Total timeline for Mechanical Team: 75 Hours

Work Breakdown Structure



KPI (Key Performance Indicator)

A measurable value that indicates how effectively the project is meeting its objectives.

Rate: 95%

Disclaimer: The KPI rate reflects the targeted performance level under planned project scope and resources. Actual outcomes may vary depending on unforeseen technical or operational challenges.

Creep (Scope Creep)

Uncontrolled or gradual expansion of project scope beyond its original objectives.

Rate: 5%

Disclaimer: The creep rate represents an estimated buffer for minor deviations. Any expansion beyond this rate will require formal change requests and client approval.

Timeline and Budget Justification

Timeline Justification

R and D

The 120-hour R&D timeline is structured to ensure a balanced and high-quality development process across hardware design, firmware implementation, and system-level validation. Significant time is allocated to schematic design and PCB layout due to the complexity of automotive power protection, high-current LED driving, and Bluetooth RF routing requirements. Firmware development receives the largest allocation to support real-time PWM control, priority-based lighting logic, Bluetooth communication, and safety fallback behavior. System integration and testing hours are reserved to validate functional correctness, noise immunity, and lighting reliability under simulated automotive conditions. The remaining time is dedicated to documentation and design optimization to ensure the final deliverables are manufacturing-ready and deployment-capable, making the timeline realistic, efficient, and aligned with automotive-grade engineering standards.

PCB Design

The 60-hour PCB design timeline is structured to ensure the development of a reliable, automotive-grade electronic design while maintaining efficiency and design quality. Significant time is allocated to schematic development and PCB layout due to the complexity of integrating automotive power protection circuits, high-current RGBW LED driving paths, and Bluetooth RF layout constraints. Additional hours are reserved for component validation, design reviews, and DFM compliance to ensure manufacturability and electrical robustness. This balanced time allocation minimizes design risks, reduces iteration cycles, and ensures the final PCB design is production-ready and compliant with automotive performance requirements.

Mechanical Design

The 75-hour mechanical design timeline is structured to ensure a robust, manufacturable, and vehicle-compatible enclosure and mounting solution for the RGBW automotive tail light system. The Concept Design phase (15 hours) allows for requirement analysis, enclosure form factor development, mounting strategy definition, and thermal airflow planning to ensure



Portfolio



Ultra-Slim Smart Health Band

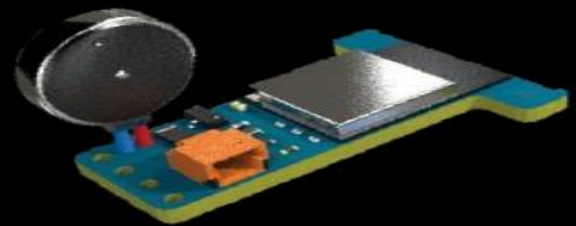
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Problem

Many wearable health devices are either bulky, power-hungry, or lack seamless connectivity and user-friendly interaction. These limitations reduce comfort, shorten battery life, and hinder reliable long-term health monitoring especially in applications such as remote patient care and continuous wellness tracking.

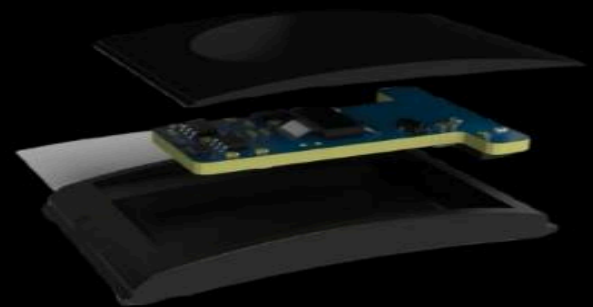


Rendered PCB

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GOALS

- Design and develop an ultra-slim, low-power smart health band for all-day wear.
- Deliver accurate biometric monitoring in a comfortable and lightweight form factor.
- Enable gesture-based, touchless control for intuitive user interaction.
- Provide haptic alerts for discreet notifications.
- Support OTA (Over-The-Air) firmware updates for long-term scalability and maintenance.
- Optimize the system for fitness, wellness, and medical IoT applications with extended battery life.



Mechanical Enclosure

Muhammad Bilal Javaid

Conclusion

This project successfully delivered an ultra-slim, low-power smart health band that meets all defined hardware, firmware, and usability objectives. The final design demonstrates reliable biometric sensing, BLE 5.0 connectivity, and gesture-based touchless interaction within a compact, comfortable form factor. With haptic feedback, OTA firmware support, and optimized power management for extended battery life, the solution is fully validated, scalable, and well-suited for fitness, wellness, and medical IoT applications.

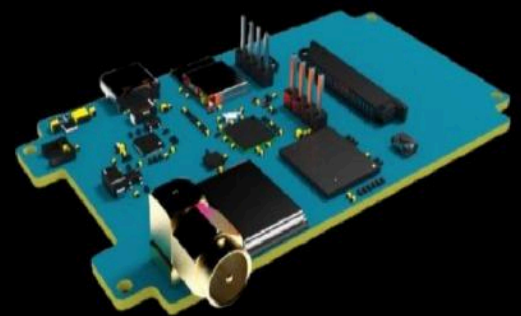
Fitness Tracking Device

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Problem

High-performance racing enthusiasts require accurate and real-time lap and section timing for performance analysis. Existing solutions often lack seamless wireless connectivity, efficient on-device processing, or reliable data transfer to companion apps, limiting usability, precision, and post-session analysis.



GOALS

- Develop a Bluetooth-enabled racing display compatible with GPS trackers such as the Crossbox CBX30 and RaceBox Mini-S.
- Enable real-time calculation and recording of lap times, section times, and session statistics directly on the device.
- Ensure seamless Bluetooth connectivity for data transfer to the Crossbox mobile app.
- Incorporate the BMD300 (nRF52832) Bluetooth module for reliable BLE connectivity.
- Design a compact, low-power PCB optimized for signal integrity, EMI control, and efficient wireless communication.
- Implement an STM32-based control architecture for stable, real-time processing and reliable device performance.



The Bluetooth-enabled racing display provides racing enthusiasts with a compact, reliable, and real-time timing solution. Its STM32-based architecture, optimized PCB design, and seamless Bluetooth connectivity allow accurate lap and section tracking, efficient on-device processing, and effortless integration with the Crossbox mobile app. This device demonstrates a robust and user-friendly approach to enhancing racing performance analysis.

Smart Glasses

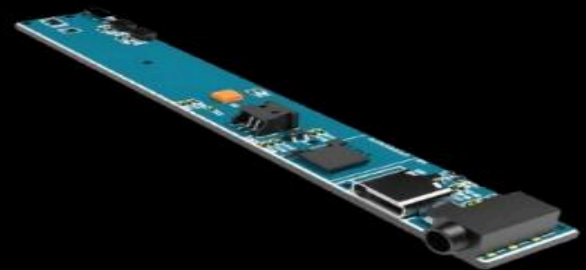
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Problem

Individuals with physical disabilities often face challenges in interacting with computers or mobile devices due to limited mobility. Traditional input devices like mice and touchscreens are not always accessible, and existing assistive solutions may lack precision, responsiveness, wireless connectivity, or efficient power management, limiting usability and convenience.



Rendered PCB

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GOALS

- Design and validate BLE-enabled smart glasses for controlling a computer or mobile device cursor using head movements.
- Integrate a compact 4-layer PCB for a robust and space-efficient hardware platform.
- Utilize the MPU9250 gyro sensor for precise head movement tracking.
- Incorporate the BMD300 (nRF52832) Bluetooth module for reliable BLE connectivity.
- Implement a Battery Management System (BMS) for efficient charging, discharging, and thermal safety.
- Develop interrupt-driven firmware to ensure seamless and responsive operation.
- Deliver an intuitive, user-friendly assistive device suitable for daily use.



Mechanical Enclosure

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Conclusion

The BLE-enabled smart glasses provide an intuitive and accessible solution for handicapped individuals, enabling precise control of computers and mobile devices through head movements. By combining a compact 4-layer PCB, MPU9250 gyro sensor, BMD300 Bluetooth module, robust Battery Management System, and responsive interrupt-driven firmware, the device ensures seamless functionality, efficient power management, and user-friendly operation. This integration demonstrates a reliable and innovative approach to assistive technology.

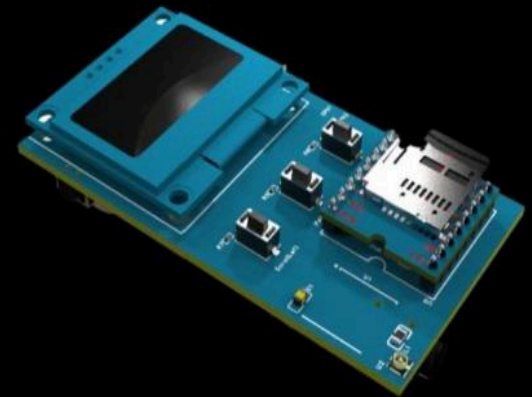
Pilgrimage Companion Device

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Problem

Pilgrims performing Tawaf, Sa'i, and Tasbeeh often face challenges in tracking rituals accurately, accessing timely guidance, and managing orientation, especially in crowded or complex environments. Existing solutions may lack real-time responsiveness, GPS-based location accuracy, bilingual support, or reliable guidance, reducing convenience and precision.



GOALS

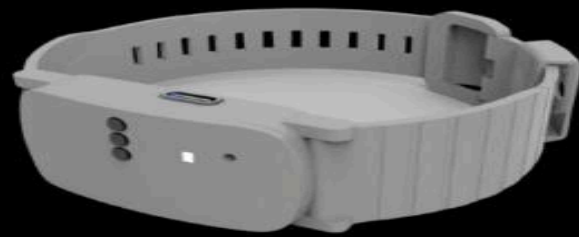
- Develop AL-Mueen, an STM32-based, event-driven pilgrimage companion.
- Integrate a SH1106 OLED display for clear visual guidance and DFPlayer audio engine for timely voice prompts
- Utilize a GNSS/GPS parser for precise location and time updates.
- Offer a bilingual experience with instant switching between English and Arabic for text and audio.
- Ensure responsive interaction, smooth UI updates, and operational reliability in demanding environments.
- Provide accurate guidance for Tawaf, Sa'i, and Tasbeeh using a cooperative state machine and interrupt-driven inputs.



AL-Mueen delivers a responsive, bilingual, and GPS-enabled pilgrimage companion that guides users through Tawaf, Sa'i, and Tasbeeh with real-time visual and audio prompts. By combining STM32-based control, GPS positioning, an OLED display, and DFPlayer audio, the device ensures accurate, user-friendly, and consistent performance, enhancing the ritual

Smart Social WristBand

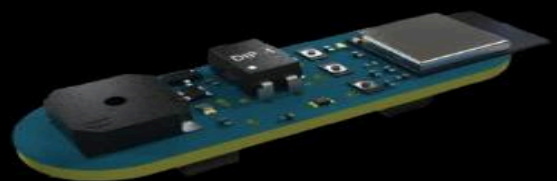
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Problem

Wearable safety and monitoring devices often suffer from short battery life, bulky hardware, limited sensor integration, and unreliable wireless connectivity. Many solutions lack a modular electronics design and flexible firmware architecture, making them difficult to adapt to different use cases or scale for future features. Additionally, poor enclosure design can compromise wearability and user comfort, reducing real-world adoption.



Render PCB

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GOALS

- Design and develop a compact, low-power wearable monitoring prototype.
- Integrate modular sensors to support multiple use cases and future expansion.
- Develop a custom PCB and BLE-based firmware for efficient data acquisition and transmission.
- Build a cross-platform mobile application for real-time monitoring and alert notifications.
- Design a wearable, ergonomic enclosure for user comfort and daily use.
- Validate the system through comprehensive testing to ensure reliability, scalability, and demo readiness.



Mechanical Enclosure

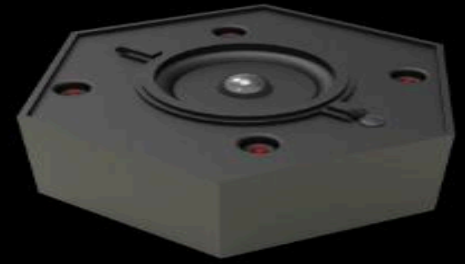
Muhammad Bilal Javaid

Conclusion

This project successfully delivered a compact, low-power wearable monitoring prototype with reliable Bluetooth Low Energy connectivity and modular sensor integration. The solution combines a custom PCB, efficient BLE firmware, and a cross-platform mobile application to enable real-time data monitoring and alerting. A carefully designed ergonomic enclosure and thorough system testing ensure user comfort, power efficiency, and dependable performance. Overall, the prototype is validated, scalable, and well-prepared for demonstration and future product deployment.

Smart Fire Safety System

Muhammad Bilal Javaid



Muhammad Bilal Javaid

Problem

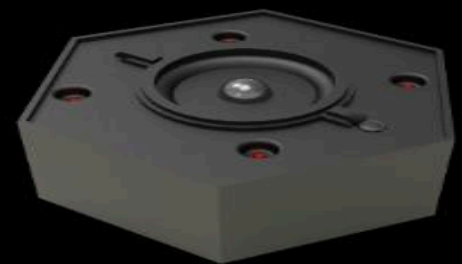
Fire hazards and poor indoor air quality pose serious risks to homes, offices, and industrial environments. Traditional fire safety systems often lack real-time monitoring, long-range communication, and intelligent alerts. Many solutions depend heavily on internet connectivity, have limited user interaction, and fail to provide early warnings for air quality degradation—leading to delayed responses and increased damage or safety risks.



Muhammad Bilal Javaid

GOALS

- Provide a reliable, wireless, and intelligent safety solution.
- Instantly detect smoke, fire, and temperature changes.
- Continuously monitor indoor air quality for early hazard detection.
- Operate reliably even without internet connectivity.
- Ensure long-range and stable communication using LoRa mesh networking
- Offer an intuitive touchscreen interface for monitoring and manual control
- Comply with EMI/EMC standards for safe and robust operation.



Mechanical Enclosure

Muhammad Bilal Javaid

Conclusion

The Smart Fire Safety and Monitoring System successfully addresses modern safety challenges by combining advanced sensors, long-range wireless communication, and an interactive user interface. With real-time alerts, offline functionality, and robust design, it provides a dependable solution for homes, offices, smart buildings, and industrial environments. This system enhances early detection, improves response time, and significantly increases overall safety and reliability.

Automated Irrigation System

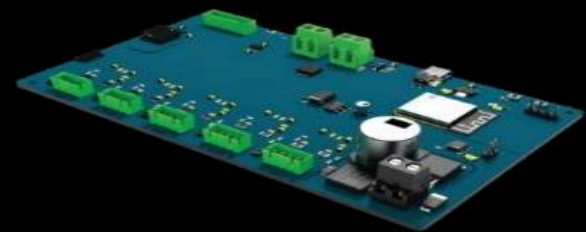
Muhammad Bilal Javaid



Muhammad Bilal Javaid

Problem

Large-scale irrigation systems often lack low-power, flexible wireless interfaces for on-site monitoring, configuration, and diagnostics. While long-distance wired communication such as RS-485 ensures sensor data reliability, the absence of short-range wireless access makes system setup, maintenance, and real-time interaction difficult in the field. Power-hungry communication methods also conflict with solar-powered deployments, limiting usability and scalability in smart farming applications.

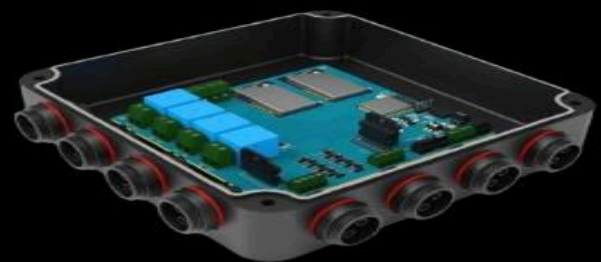


Rendered PCB

Muhammad Bilal Javaid

GOALS

- Design and develop an automatic, high-precision dye dispensing system with visual feedback and wireless control.
- Create a flexible LED PCB to display fluid color in real time during operation.
- Integrate Arduino-based control for multiple peristaltic pumps.
- Use NEMA 17 stepper motors with TMC2209 drivers for accurate and smooth liquid dispensing.
- Enable Bluetooth Low Energy (BLE) connectivity for mobile app-based monitoring and remote control.
- Ensure the system can handle higher power requirements and operate reliably under varying loads.
- Achieve consistent, repeatable, and precise dye dispensing for industrial use.



Mechanical Enclosure

Muhammad Bilal Javaid

Conclusion

This project successfully delivers a smart irrigation solution that leverages BLE for efficient, low-power wireless access alongside RS-485 for stable long-distance sensor communication. BLE enables seamless mobile-based monitoring, control, and system configuration, significantly improving usability during installation and maintenance. Combined with solar-powered operation, weatherproof hardware, and centralized pump control, the system provides a scalable, energy-efficient, and user-friendly platform for precision agriculture, remote farm management, and water conservation initiatives.

Client Feedback

Technology Officer for Electronic Product Development

Great to work with Muhammad and his team



Project Budget: \$4,550

Pcb designer

Very professional team.



Project Budget: \$5,250

PCB Layout

Muhammad and his team executed the project flawlessly with clear communication, quick responses, and efficient progress tracking. They handled changes smoothly and consistently aimed for fast solutions. Highly recommended for future projects.



Project Budget: \$12,000

CAD design- Prototype

Muhammad and his team managed the project flawlessly from start to finish, with clear communication, fast responses, and excellent transparency via Google Docs. They efficiently handled changes and consistently delivered quick solutions. The client highly recommends them and would gladly work with them again.



Project Budget: \$3,500

PCB and Electrical design for a photoreactor

I enjoy working with this group. They are talented.



Project Budget: \$15,324

Electrical Engineer - IoT Kapton Heater Controller

Bilal and his team did good work. They are quick and communicate well, and are accommodating for any issues.



Project Budget: \$3,658

ESP32 Developer Needed for Feature Enhancement

Excellent team. They are a decent sized team with lot of PCB development experience and firmware development. I worked with a team before and this team has exceeded my expectations. Will use again when I need. Thanks for all the help and assistance.



Project Budget: \$8,759

Arduino/RC32 Developer for GPS Based Alert System for Airsoft Game

It was a pleasure working with this team. Great service and product!



Project Budget: \$5,823

Advance Existing Prototype - for Ultra Small Smart Flowmeter

Muhammad and his team delivered high-quality work by downsizing the PCB and choosing excellent components. Although firmware delivery took longer than expected, their accommodating nature and ability to bring in expert help led to a successful and satisfying project outcome.



Project Budget: \$18,000

PCB Layout

Muhammad and his team managed the project flawlessly from start to finish, with clear communication, quick responses, and transparent progress tracking. They handled changes efficiently and delivered fast solutions. Highly recommended for future collaborations.



Project Budget: \$5,000

Smart Alarm Clock With E-ink Display

Fast response time, good planning from the start and good support. It was a pleasure to work with the team, and I hope to rehire soon again.



Project Budget: \$3,000

Help Build Supplement Vending Machine Prototype

Muhammad Bilal and his team are very diligent, creative and hard working. their ability to be able to problem solve and create solutions for work moving forward has enabled the success of our project now and for the future. I would recommend to anyone.



Project Budget: \$ 5,000

Schematic Design for Arduino Based Project with Stepper Motors

Exceptional product and service



Project Budget: \$17,312

Automated Electric Shaver Development for Back Neck/Hairline

The team is extremely responsive and thorough. Highly recommend



Project Budget: \$3,658

Mechanical Engineer Needed for Steel Bullet Trap Design with Electronics Integration

Highly skilled and responsive, delivering quality work with great attention to detail.



Project Budget: \$14,310

Thank You!

Thank you for your time and consideration. We would be happy to discuss our proposal further, address any questions you may have, and provide additional insights.

We look forward to your feedback and the opportunity to collaborate on the next steps.

CEO

MUHAMMAD BILAL JAVAID

200+

Clients.

100+

Project.

30+

Countries.

98%

Response rate